High Tech Marketing Text Analysis

Brand: Motorola Razr 2025

Group 12?

Introduction

Explain shit here philippe 8=====D OHMAGHADIMBOUTTABUUUUSSSTTTTT

Sample Selection

We decided to go with mulitiple sources of text in an attempt to capture multiple perspectives on the Razr and avoid pigeon hollling into one corner of the internet which may have skewed results.

Data Sources

We grabbed our shit from Reddit threads about the Motorola Razr 2025 (pain in my ass) and Youtube comments from popular tech youtubers who reviewed the Razr 2025

Sampling representation & Bias

Explain if the sample is representative of all customers, and how platform choice may skew results e.g Reddit chuds are not normal people.

Data Collection

```
fetch_reddit_thread <- function(thread_url, limit = 500, attempts = 5) {
  api_url <- paste0(thread_url, ".json?limit=", limit)
  response <- RETRY(
    "GET",
    api_url,
    user_agent(paste0("Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) ",
                      "AppleWebKit/537.36 (KHTML, like Gecko) ",
                      "Chrome/125.0.0.0 Safari/537.36")),
    times        = attempts,
    pause_base   = 2,
    pause_cap    = 20,
    terminate_on = c(400, 401, 403, 404)
  )
```

```r
if (http_error(response)) {
  stop("HTTP ", status_code(response), " when fetching ", api_url)
}

payload <- fromJSON(
  content(response, as = "text", encoding = "UTF-8"),
  simplifyVector = FALSE
)
post <- payload[[1]]$data$children[[1]]$data

flatten_comments <- function(children) {
  if (!is.list(children)) return(data.frame())
  rows <- list()
  for (child in children) {
    if (!is.null(child$kind) && child$kind == "t1") {
      comment   <- child$data
      body_text <- comment$body %||% NA_character_
      if (!is.null(body_text)) body_text <- gsub("[\r\n]+", " ", body_text)
      rows[[length(rows) + 1]] <- data.frame(
        type      = "comment",
        id        = comment$id        %||% NA_character_,
        parent_id = comment$parent_id %||% NA_character_,
        author    = comment$author    %||% NA_character_,
        body      = body_text,
        score     = comment$score     %||% NA_real_,
        subreddit = comment$subreddit %||% NA_character_,
        stringsAsFactors = FALSE
      )
      replies <- if (is.list(comment$replies) &&
                     !is.null(comment$replies$data$children))
        comment$replies$data$children else list()
      if (length(replies) > 0)
        rows <- c(rows, flatten_comments(replies))
    }
  }
  if (length(rows) == 0) return(data.frame())
  rows <- Filter(function(x) is.data.frame(x) && ncol(x) > 0, rows)
  if (length(rows) == 0) return(data.frame())
  tryCatch(
    dplyr::bind_rows(rows),
    error = function(e) {
      cols     <- c("type","id","parent_id","author","body","score","subreddit")
      repaired <- lapply(rows, function(r) {
        if (!is.data.frame(r)) return(NULL)
        for (m in setdiff(cols, names(r))) r[[m]] <- NA
        r[cols]
      })
      dplyr::bind_rows(Filter(Negate(is.null), repaired))
    }
  )
}

comments  <- flatten_comments(payload[[2]]$data$children %||% list())
```

```
  post_body <- post$selftext %||% post$title %||% NA_character_
  if (!is.null(post_body)) post_body <- gsub("[\r\n]+", " ", post_body)

  post_df <- data.frame(
    type      = "post",
    id        = post$id     %||% NA_character_,
    parent_id = NA_character_,
    author    = post$author %||% NA_character_,
    body      = post_body,
    score     = post$score  %||% NA_real_,
    subreddit = post$subreddit %||% NA_character_,
    stringsAsFactors = FALSE
  )

  if (!is.data.frame(comments)) comments <- data.frame()
  tryCatch(
    dplyr::bind_rows(post_df, comments),
    error = function(e) post_df
  )
}

suppressWarnings({
  reddit_cache <- "data/reddit_content.csv"
  cached <- file.exists(reddit_cache)

  if (cached) {
    reddit_all <- tryCatch(
      read.csv(reddit_cache, stringsAsFactors = FALSE),
      error = function(e) { message("Cache read failed: ", conditionMessage(e)); data.frame() }
    )
  } else {
    search_config <- list(
      list(sub = "Android",       kw = "motorola razr"),
      list(sub = "Android",       kw = "razr 2025"),
      list(sub = "Android",       kw = "razr flip"),
      list(sub = "Motorola",      kw = "razr"),
      list(sub = "Motorola",      kw = "razr ultra"),
      list(sub = "Motorola",      kw = "razr plus"),
      list(sub = "phones",        kw = "razr"),
      list(sub = "phones",        kw = "motorola"),
      list(sub = "smartphones",   kw = "motorola razr"),
      list(sub = "razr",          kw = "razr"),
      list(sub = "razr",          kw = "motorola"),
      list(sub = "technology",    kw = "motorola razr"),
      list(sub = "technology",    kw = "razr 2025")
    )

    reddit_posts <- dplyr::bind_rows(Filter(
      function(x) is.data.frame(x) && nrow(x) > 0,
      lapply(search_config, function(cfg) {
        result <- tryCatch({
          Sys.sleep(2)
          find_thread_urls(keywords = cfg$kw, subreddit = cfg$sub, sort_by = "top")
```

```
        }, error = function(e) {
          message("Failed r/", cfg$sub, " / '", cfg$kw, "': ", conditionMessage(e))
          data.frame(url = character())
        })
        if (!is.data.frame(result)) return(data.frame(url = character()))
        result
      })
    )) |> dplyr::distinct(url, .keep_all = TRUE)

    reddit_max_threads <- 250

    if (nrow(reddit_posts) > 0) {
      to_fetch <- head(reddit_posts$url, reddit_max_threads)
      raw_list <- lapply(seq_along(to_fetch), function(i) {
        Sys.sleep(1)
        tryCatch(
          fetch_reddit_thread(to_fetch[i]),
          error = function(e) { message("Failed: ", conditionMessage(e)); data.frame() }
        )
      })
      reddit_all <- dplyr::bind_rows(
        Filter(function(x) is.data.frame(x) && nrow(x) > 0, raw_list)
      )
    } else {
      reddit_all <- data.frame()
    }

    if (!dir.exists("data")) dir.create("data")
    write.csv(reddit_all, reddit_cache, row.names = FALSE)
  }
})

cat("Reddit total rows:", nrow(reddit_all), "\n")
```

```
## Reddit total rows: 992
```

```
cat("Posts vs comments:\n");  print(table(reddit_all$type))
```

```
## Posts vs comments:

##
## comment    post
##     765     227
```

```
cat("By subreddit:\n");       print(table(reddit_all$subreddit))
```

```
## By subreddit:

##
##     Android    motorola      phones        razr Smartphones
##           6         312          29         608          37
```

```r
fetch_youtube_comments <- function(video_id, max_comments = 400, api_key = Sys.getenv("YOUTUBE_KEY")) {
  if (identical(api_key, "")) {
    stop("YOUTUBE_KEY is not set. Load it from .env or your environment before knitting.")
  }

  all_comments <- list()
  page_token   <- NULL

  repeat {
    query <- list(
      part       = "snippet",
      videoId    = video_id,
      maxResults = 100,
      textFormat = "plainText",
      key        = api_key
    )
    if (!is.null(page_token)) query$pageToken <- page_token

    resp <- httr::GET("https://www.googleapis.com/youtube/v3/commentThreads",
                      query = query)

    if (httr::http_error(resp)) {
      error_body <- tryCatch(httr::content(resp, as = "text", encoding = "UTF-8"),
                             error = function(e) "")
      message("YouTube API error: ", httr::status_code(resp),
              if (nzchar(error_body)) paste0(" - ", error_body) else "")
      break
    }

    parsed <- jsonlite::fromJSON(
      httr::content(resp, as = "text", encoding = "UTF-8"),
      flatten = TRUE
    )

    items <- parsed$items
    if (is.null(items) || nrow(items) == 0) break

    batch <- data.frame(
      comment_id = items$id,
      author     = items$snippet.topLevelComment.snippet.authorDisplayName,
      body       = items$snippet.topLevelComment.snippet.textDisplay,
      like_count = items$snippet.topLevelComment.snippet.likeCount,  # kept for segmentation
      stringsAsFactors = FALSE
    )

    all_comments[[length(all_comments) + 1]] <- batch
    collected <- sum(sapply(all_comments, nrow))

    page_token <- parsed$nextPageToken
    if (is.null(page_token) || collected >= max_comments) break
    Sys.sleep(0.5)
  }
```

```
  if (length(all_comments) == 0) return(data.frame())
  dplyr::bind_rows(all_comments) |> dplyr::mutate(source = "youtube")
}

razr_videos <- list(
  mkbhd      = "8om1eJrO2lU",   # MKBHD Razr 2025 review
  dave2d     = "JTT67FSc8j8",   # ShortCircuit review
  mrmobile   = "YGTkjchlVJk",   # MrMobile review
  phonearena = "b3Ijuf7DHcQ"    # PhoneArena review
)

if (file.exists("data/youtube_comments.csv")) {
  youtube_all <- read.csv("data/youtube_comments.csv", stringsAsFactors = FALSE)
  cat("Loaded YouTube comments from CSV:", nrow(youtube_all), "\n")
} else {
  youtube_all <- purrr::map2_dfr(
    razr_videos,
    names(razr_videos),
    ~ fetch_youtube_comments(.x, max_comments = 400) |>
        dplyr::mutate(channel = .y)
  )
  if (!dir.exists("data")) dir.create("data")
  write.csv(youtube_all, "data/youtube_comments.csv", row.names = FALSE)
}
```

```
## Loaded YouTube comments from CSV: 1161
```

```
cat("YouTube comments collected:", nrow(youtube_all), "\n")
```

```
## YouTube comments collected: 1161
```

```
print(table(youtube_all$channel))
```

```
##
##     dave2d      mkbhd   mrmobile phonearena
##        400        400        257        104
```

Exploration

Lets have a gander at the data we collected and see if theres any thing fun before we clean it up.

```
exploration_summary <- tibble::tibble(
  Source    = c("Reddit (comments only)", "YouTube"),
  Documents = c(sum(reddit_all$type == "comment", na.rm = TRUE),
                nrow(youtube_all)),
  Avg_chars = c(round(mean(nchar(reddit_all$body[reddit_all$type == "comment"]), na.rm = TRUE), 1),
                round(mean(nchar(youtube_all$body), na.rm = TRUE), 1))
)
knitr::kable(exploration_summary, caption = "Raw corpus volume by source")
```

Table 1: Raw corpus volume by source

Source	Documents	Avg_chars
Reddit (comments only)	765	218.1
YouTube	1161	153.8

Pre-Processing

we have 2 separate data frames for reddit, and youtube, we need to merge them into one big one for the text analysis. We can add a column to indicate the source of each row and then bind them together.

```
##
##      de      en      es      fr      pt      ro      ru xx-Qaai
##       1    1792       7       1       2       1       1       1
```

Side note maybe we dont have to do all of them based on the quality of the data we get, but we should at least explain how we would do them and why they are important.

Filtering & Cleaning

```
# Keep English only
combined_data <- combined_data |> filter(language == "en")

# Strip non-ASCII characters (emoji, accents that break the DFM)
combined_data$body <- gsub("[^\x01-\x7F]", "", combined_data$body)

# Remove now-empty bodies
combined_data <- combined_data |> filter(body != "")

# --- Old vs. New era flag (brand revitalisation comparison) ---
# Flag documents that explicitly reference the original/heritage Razr.
old_terms <- c("original", "old razr", "2004", "2005", "v3",
               "flip phone days", "back in the day", "nostalgia",
               "first razr", "classic", "used to have", "the old one")
combined_data <- combined_data |>
  mutate(era = ifelse(
    grepl(paste(old_terms, collapse = "|"), body, ignore.case = TRUE),
    "references_old", "new_only"))

cat("Era distribution (old-brand references vs new-only):\n")
```

```
## Era distribution (old-brand references vs new-only):
```

```
print(table(combined_data$era))
```

```
##
##       new_only references_old
##           1753             39
```

```r
cat("\nDocuments by platform:\n")
```

```
##
## Documents by platform:
```

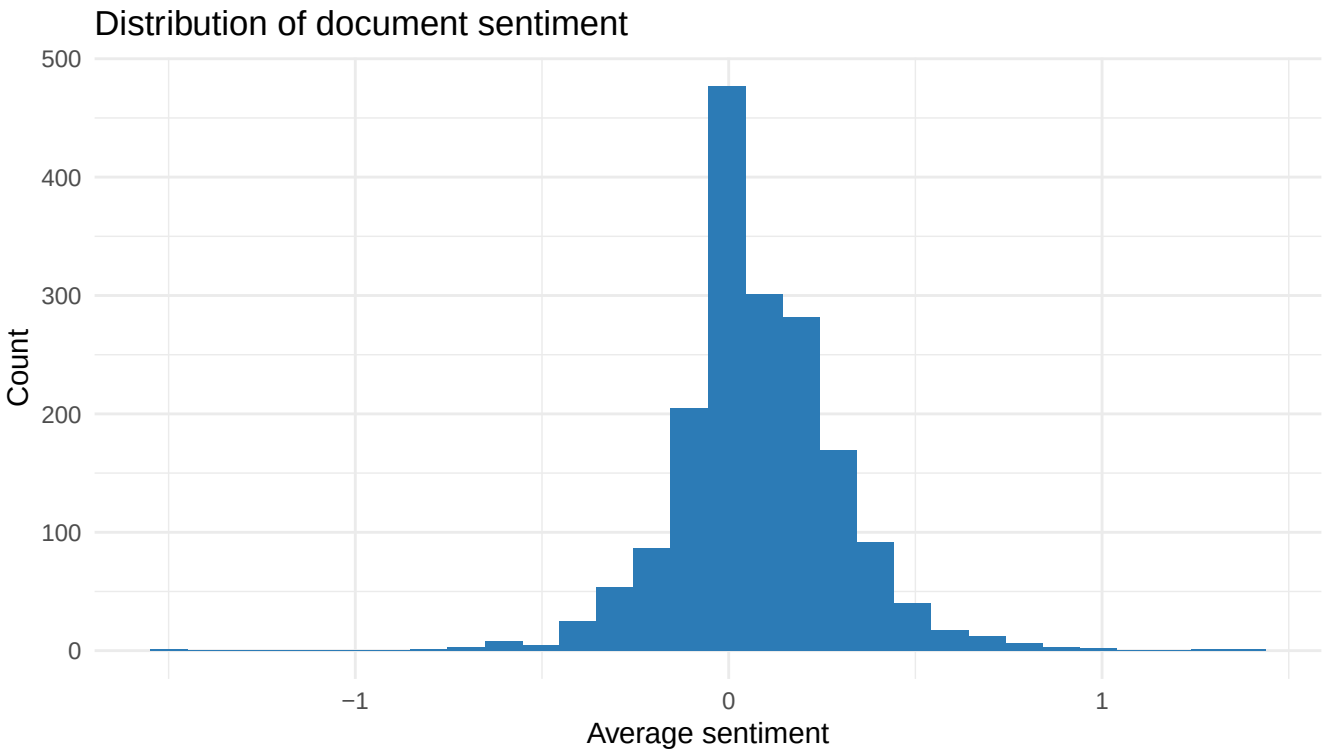
```r
print(table(combined_data$platform))
```

```
##
##  Reddit YouTube
##     682    1110
```

Modeling

Corpus Creation

```r
corpus <- quanteda::corpus(combined_data, docid_field = "id", text_field = "body")

# Attach document-level sentiment (used later for topic x sentiment analysis)
quanteda::docvars(corpus, field = "ave_sentiment") <-
  sentimentr::sentiment_by(sentimentr::get_sentences(corpus))$ave_sentiment

# Distribution of sentiment across the corpus
ggplot(data.frame(s = quanteda::docvars(corpus)$ave_sentiment),
       aes(x = s)) +
  geom_histogram(bins = 30, fill = "#2C7BB6") +
  labs(title = "Distribution of document sentiment",
       x = "Average sentiment", y = "Count") +
  theme_minimal()
```



Tokenization

Explain how we tokenized the data and why.

```
tokens <- quanteda::tokens(corpus,
                           what = "word",
                           remove_punct = TRUE,
                           remove_symbols = TRUE,
                           remove_numbers = TRUE,
                           remove_url = TRUE,
                           remove_separators = TRUE,
                           split_hyphens = FALSE,
                           split_tags = FALSE,
                           include_docvars = TRUE,
                           padding = FALSE,
                           verbose = FALSE)

tokens <- quanteda::tokens_tolower(tokens)
```

Synonym consolidation

```
tokens <- quanteda::tokens_replace(tokens,
  pattern = c(
    "foldable", "folding",
    "flipping",
    "cam", "cams",
```

```
    "charging", "charger",
    "displays",
    "hinges", "hinging",
    "pricing", "priced", "pricey", "expensive", "costly",
    "galaxy",
    "ios", "apple"
  ),
  replacement = c(
    "fold", "fold",
    "flip",
    "camera", "camera",
    "charge", "charge",
    "display",
    "hinge", "hinge",
    "price", "price", "price", "price", "price",
    "samsung",
    "apple", "apple"
  ),
  valuetype = "fixed")
```

Stop Words Removal

```
custom_stops <- c(
  "razr", "motorola", "phone", "just", "get", "like", "thing", "really",
  "know", "time", "people", "would", "also", "much", "use", "see",
  "one", "will", "can", "even", "still", "want", "review",
  "moto", "device", "smartphone", "im", "dont", "ive", "thats", "got",
  "amp", "u", "ur", "ok", "yeah", "lol", "gonna", "good", "make",
  "think", "need", "look", "come", "say"
)
tokens <- quanteda::tokens_remove(
  tokens, pattern = c(quanteda::stopwords("en"), custom_stops))
```

Stemming & Lemmatization

```
feats <- quanteda::types(tokens)
tokens <- quanteda::tokens_replace(
  tokens,
  pattern     = feats,
  replacement = textstem::lemmatize_words(feats),
  valuetype   = "fixed")
```

DFM Creation & Frequency Filtering

```
dfm <- quanteda::dfm(tokens)

# Keep only tokens of 3+ characters (drops residual "u", "id", "ui", etc.)
```

```r
dfm <- quanteda::dfm_select(dfm, pattern = "^.{3,}$",
                            valuetype = "regex", selection = "keep")

cat("Features before frequency trim:", nfeat(dfm), "\n")
```

```
## Features before frequency trim: 4052
```

```r
# Drop very rare terms (appear fewer than 10 times across the whole corpus)
dfm <- dfm_trim(dfm, min_termfreq = 10)
cat("Features after frequency trim:", nfeat(dfm), "\n")
```

```
## Features after frequency trim: 559
```

```r
cat("\nMost frequent features:\n")
```

```
##
## Most frequent features:
```

```r
print(topfeatures(dfm, 15))
```

```
##  screen    fold    good    flip   phone   ultra samsung    year     app     buy
##     424     265     242     215     201     201     181     179     173     168
##     use  camera    love    work     now
##     164     162     156     138     136
```

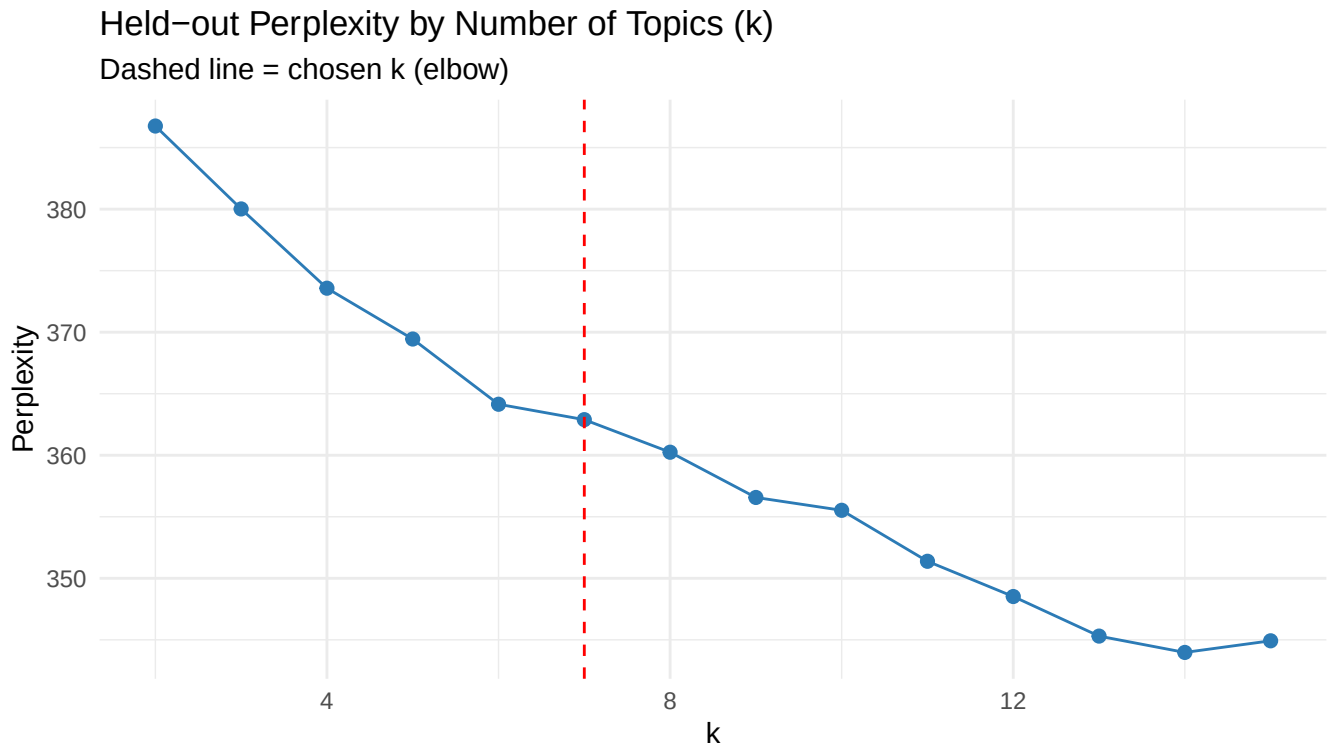
```r
# Drop any documents left empty after all filtering
empty_docs <- which(rowSums(dfm) == 0)
if (length(empty_docs) > 0) dfm <- dfm[-empty_docs, ]
cat("\nFinal DFM:", ndoc(dfm), "documents x", nfeat(dfm), "features\n")
```

```
##
## Final DFM: 1737 documents x 559 features
```

Selecting K

```r
dtm <- quanteda::convert(dfm, to = "topicmodels")

set.seed(42)
n_docs    <- nrow(dtm)
train_idx <- sample(n_docs, floor(n_docs * 0.8))
dtm_train <- dtm[train_idx, ]
dtm_test  <- dtm[-train_idx, ]

k_values <- 2:15
perplexity_scores <- sapply(k_values, function(k) {
  m <- topicmodels::LDA(dtm_train, k = k, method = "Gibbs",
                        control = list(seed = 42, iter = 1000, burnin = 200))
  topicmodels::perplexity(m, newdata = dtm_test)
```

```
})

perplexity_df <- data.frame(k = k_values, perplexity = perplexity_scores) |>
  mutate(improvement = dplyr::lag(perplexity) - perplexity)

ggplot(perplexity_df, aes(x = k, y = perplexity)) +
  geom_line(colour = "#2C7BB6") +
  geom_point(colour = "#2C7BB6", size = 2) +
  geom_vline(xintercept = 7, linetype = "dashed", colour = "red") +
  labs(title = "Held-out Perplexity by Number of Topics (k)",
       subtitle = "Dashed line = chosen k (elbow)",
       x = "k", y = "Perplexity") +
  theme_minimal()
```




```
knitr::kable(perplexity_df, digits = 2,
             caption = "Held-out perplexity and per-step improvement by k")
```

Table 2: Held-out perplexity and per-step improvement by k

k	perplexity	improvement
2	386.75	NA
3	380.02	6.74
4	373.57	6.44
5	369.44	4.13
6	364.14	5.30

k	perplexity	improvement
7	362.89	1.25
8	360.25	2.65
9	356.57	3.68
10	355.53	1.04
11	351.38	4.14
12	348.52	2.86
13	345.30	3.22
14	343.97	1.33
15	344.92	-0.95

LDA

```
K <- 7

set.seed(42)
lda_model <- seededlda::textmodel_lda(dfm, k = K)

# Assign each document its most probable topic, with its sentiment + metadata
topic_assignments <- data.frame(
  doc_id    = docnames(dfm),
  topic     = topics(lda_model),
  sentiment = docvars(dfm)$ave_sentiment,
  stringsAsFactors = FALSE
)
```

Stability Analysis

```
seeds <- c(42L, 123L, 999L)

stability_results <- lapply(seeds, function(s) {
  set.seed(s)
  m     <- seededlda::textmodel_lda(dfm, k = K)
  trms  <- seededlda::terms(m, 5)              # 5 rows (words) x K cols (topics)
  apply(trms, 2, paste, collapse = ", ")       # one string per topic
})

stability_df <- data.frame(
  topic    = paste0("Topic ", 1:K),
  seed_42  = stability_results[[1]],
  seed_123 = stability_results[[2]],
  seed_999 = stability_results[[3]],
  row.names = NULL
)

knitr::kable(stability_df,
             col.names = c("Topic", "Seed 42", "Seed 123", "Seed 999"),
             caption   = "Top 5 words per topic across 3 seeds (stability check)")
```

Table 3: Top 5 words per topic across 3 seeds (stability check)

Topic	Seed 42	Seed 123	Seed 999
Topic 1	screen, fold, case, now, year	year, iphone, android, update, since	year, iphone, samsung, now, update
Topic 2	fold, mine, wait, day, say	charge, video, now, battery, call	app, use, work, mode, google
Topic 3	flip, screen, small, phone, love	screen, small, fold, outside, big	screen, case, break, issue, work
Topic 4	samsung, year, iphone, android, phone	flip, samsung, phone, love, fold	fold, wait, mine, come, say
Topic 5	good, price, ultra, battery, model	fold, case, wait, mine, come	charge, camera, video, battery, call
Topic 6	camera, app, use, call, play	good, price, ultra, model, great	flip, love, phone, screen, small
Topic 7	app, update, display, good, screen	app, camera, try, work, google	good, ultra, price, samsung, model

Results

Topic Labels

```
# Top 10 terms per topic
topic_terms <- seededlda::terms(lda_model, 10)
knitr::kable(topic_terms, caption = "Top 10 terms per topic")
```

Table 4: Top 10 terms per topic

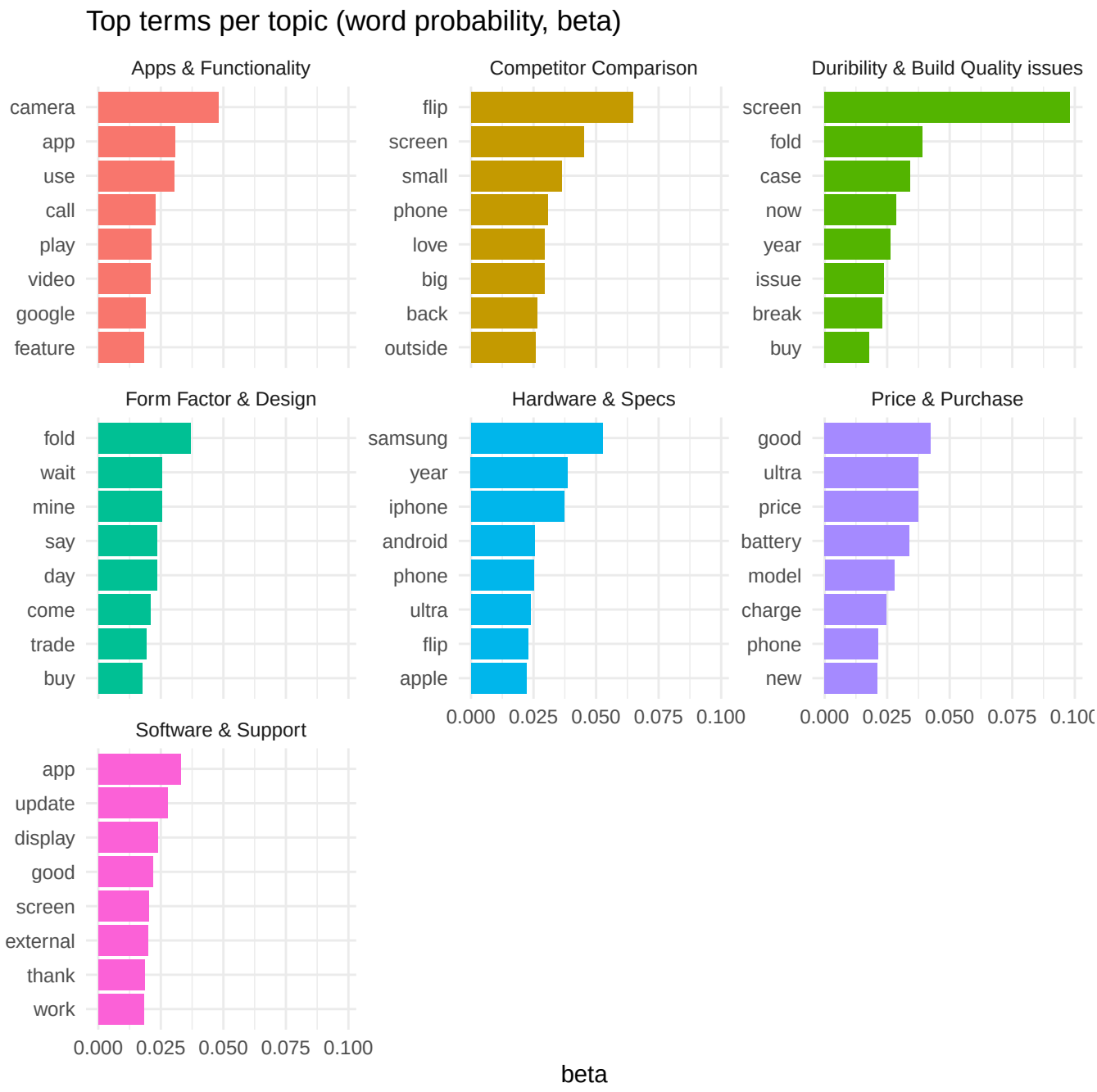
topic1	topic2	topic3	topic4	topic5	topic6	topic7
screen	fold	flip	samsung	good	camera	app
fold	mine	screen	year	price	app	update
case	wait	small	iphone	ultra	use	display
now	day	phone	android	battery	call	good
year	say	love	phone	model	play	screen
issue	come	big	ultra	charge	video	external
break	trade	back	flip	phone	google	thank
buy	buy	outside	apple	new	feature	work
drop	ship	fold	love	plus	watch	use
little	deal	hand	good	great	pixel	try

```
# Human-readable labels - REVIEW these against the table above and adjust.
topic_labels <- c(
  "topic1" = "Duribility & Build Quality issues",
  "topic2" = "Form Factor & Design",
  "topic3" = "Competitor Comparison",
  "topic4" = "Hardware & Specs",
  "topic5" = "Price & Purchase",
  "topic6" = "Apps & Functionality",
```

```
  "topic7" = "Software & Support"
)
```

Topic Word Distributions

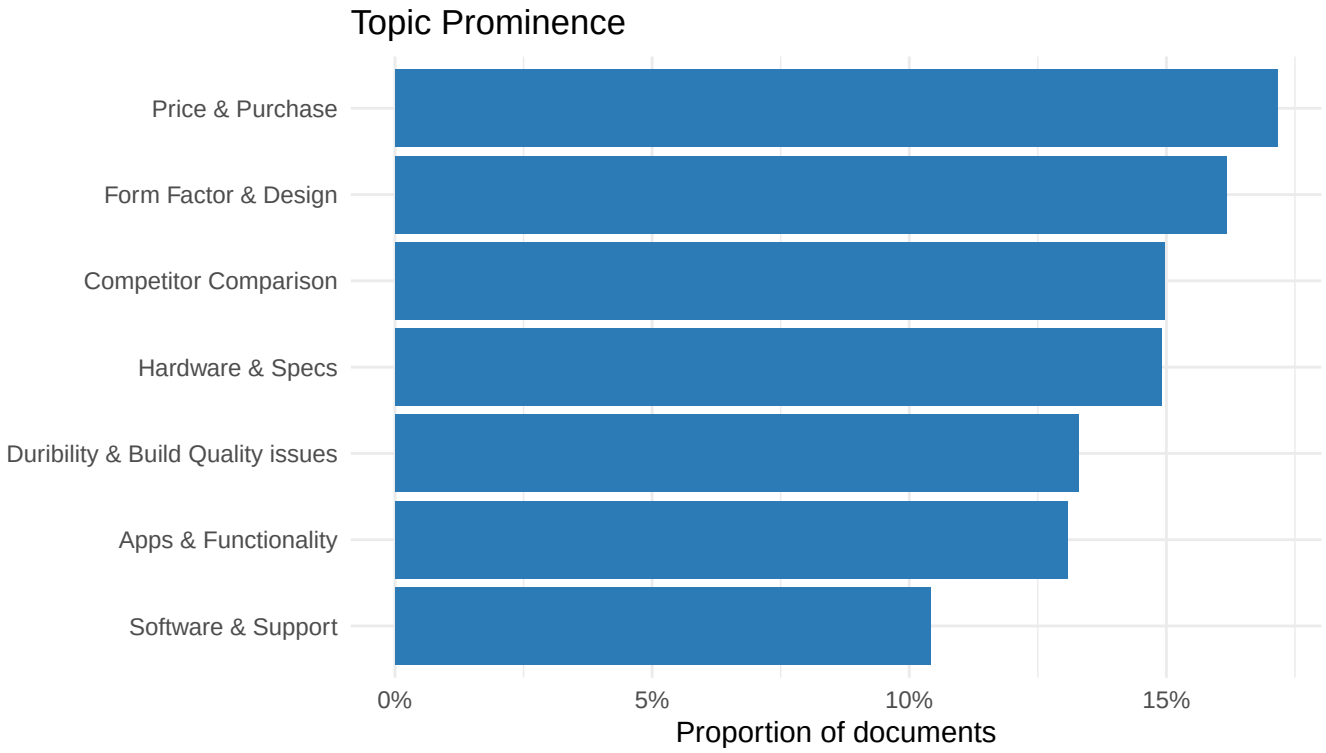
```
# Extract per-topic word probabilities (phi) from the seededlda model
phi <- lda_model$phi                        # rows = topics, cols = terms
beta_df <- as.data.frame(t(phi)) |>
  tibble::rownames_to_column("term") |>
  tidyr::pivot_longer(-term, names_to = "topic", values_to = "beta") |>
  group_by(topic) |>
  slice_max(beta, n = 8, with_ties = FALSE) |>
  ungroup() |>
  mutate(label = topic_labels[topic])

ggplot(beta_df,
       aes(x = tidytext::reorder_within(term, beta, topic),
           y = beta, fill = label)) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~label, scales = "free_y") +
  coord_flip() +
  tidytext::scale_x_reordered() +
  labs(title = "Top terms per topic (word probability, beta)",
       x = NULL, y = "beta") +
  theme_minimal()
```



Topic Prominence

```
topic_counts <- as.data.frame(table(topics(lda_model))) |>
  rename(topic = Var1, n = Freq) |>
  mutate(prop  = n / sum(n),
         label = topic_labels[as.character(topic)])

ggplot(topic_counts, aes(x = reorder(label, prop), y = prop)) +
  geom_col(fill = "#2C7BB6") +
```

```
  coord_flip() +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Topic Prominence",
       x = NULL, y = "Proportion of documents") +
  theme_minimal()
```



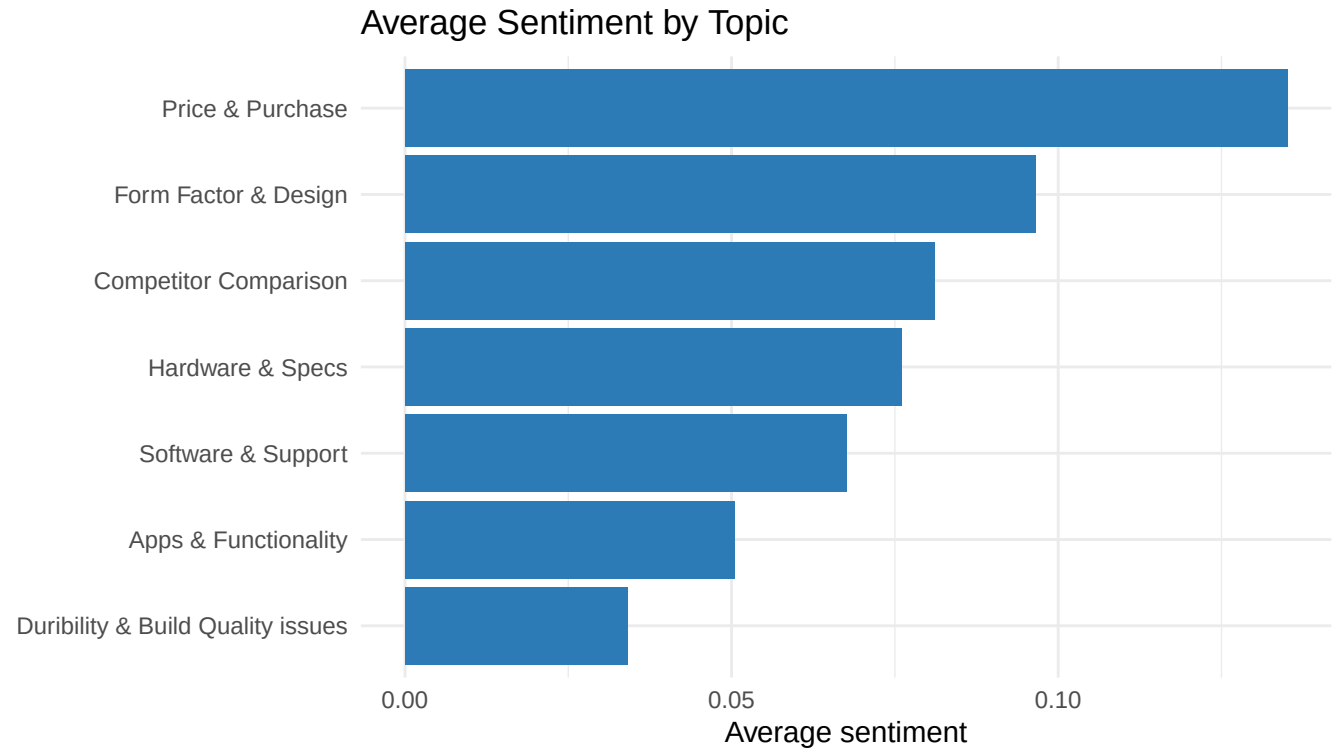

```
knitr::kable(topic_counts |> select(label, n, prop),
             digits = 3, col.names = c("Topic", "N", "Proportion"),
             caption = "Relative prominence of each topic")
```

Table 5: Relative prominence of each topic

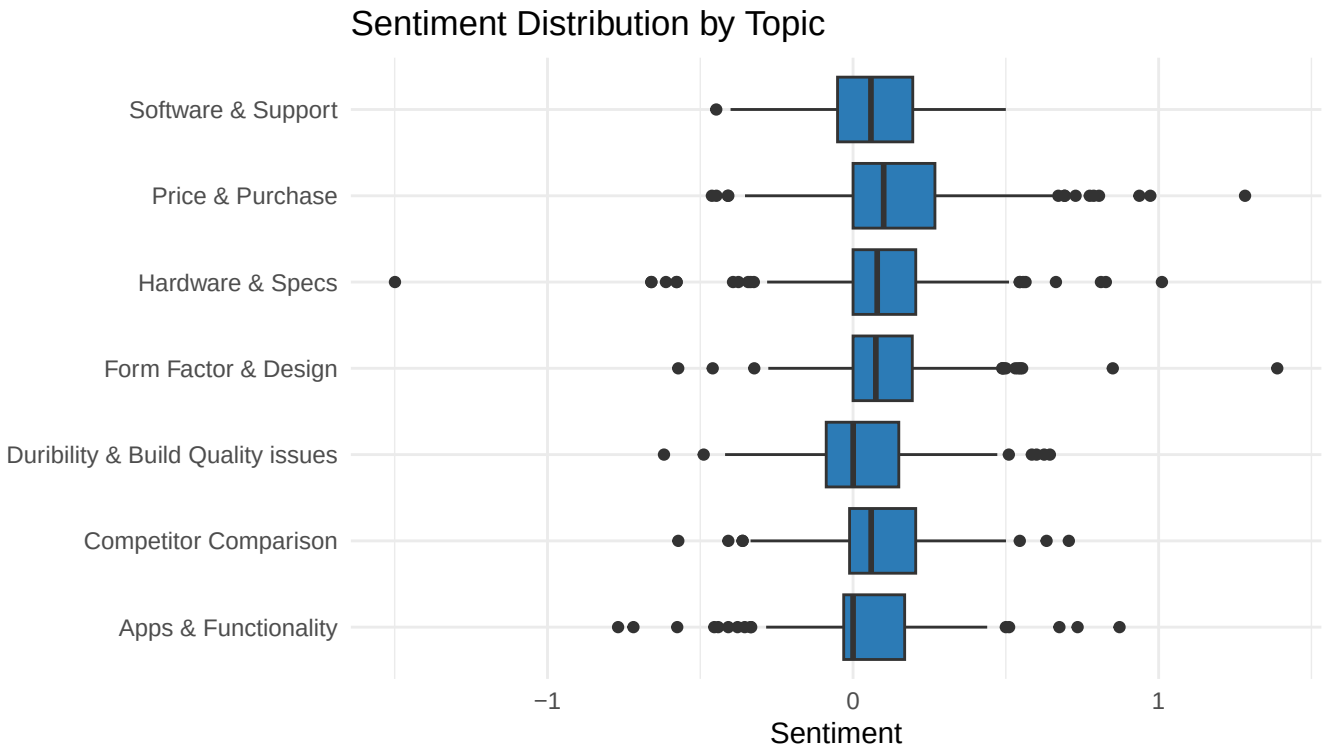
Topic	N	Proportion
Duribility & Build Quality issues	231	0.133
Form Factor & Design	281	0.162
Competitor Comparison	260	0.150
Hardware & Specs	259	0.149
Price & Purchase	298	0.172
Apps & Functionality	227	0.131
Software & Support	181	0.104

Sentiment by Topic

```r
topic_sentiment <- topic_assignments |>
  group_by(topic) |>
  summarise(mean_sentiment = mean(sentiment, na.rm = TRUE),
            n = n(), .groups = "drop") |>
  mutate(label = topic_labels[topic])

ggplot(topic_sentiment, aes(x = reorder(label, mean_sentiment), y = mean_sentiment)) +
  geom_col(fill = "#2C7BB6") +
  coord_flip() +
  labs(title = "Average Sentiment by Topic",
       x = NULL, y = "Average sentiment") +
  theme_minimal()
```




```r
ggplot(topic_assignments |> mutate(label = topic_labels[topic]),
       aes(x = label, y = sentiment)) +
  geom_boxplot(fill = "#2C7BB6") +
  coord_flip() +
  labs(title = "Sentiment Distribution by Topic",
       x = NULL, y = "Sentiment") +
  theme_minimal()
```



Statistical Tests

Do Reddit and YouTube discuss different topics?

```
topic_source <- topic_assignments |>
  mutate(doc_id = as.integer(doc_id)) |>
  left_join(combined_data |> select(id, platform, era),
            by = c("doc_id" = "id")) |>
  mutate(label = topic_labels[topic])

contingency <- table(topic_source$label, topic_source$platform)
knitr::kable(contingency, caption = "Topic distribution: Reddit vs YouTube")
```

Table 6: Topic distribution: Reddit vs YouTube

	Reddit	YouTube
Apps & Functionality	85	148
Competitor Comparison	32	223
Duribility & Build Quality issues	95	135
Form Factor & Design	179	104
Hardware & Specs	70	185
Price & Purchase	108	189
Software & Support	101	83

```r
chi_result <- chisq.test(contingency)
print(chi_result)
```

```
##
##  Pearson's Chi-squared test
##
## data:  contingency
## X-squared = 181.38, df = 6, p-value < 2.2e-16
```

Does sentiment differ across topics?

```r
kw_result <- kruskal.test(sentiment ~ factor(topic), data = topic_assignments)
print(kw_result)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  sentiment by factor(topic)
## Kruskal-Wallis chi-squared = 29.963, df = 6, p-value = 3.994e-05
```

```r
# Post-hoc pairwise comparison (Benjamini-Hochberg corrected)
pw <- pairwise.wilcox.test(topic_assignments$sentiment,
                           topic_assignments$topic,
                           p.adjust.method = "BH")
print(pw)
```

```
##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
##
## data:  topic_assignments$sentiment and topic_assignments$topic
##
##        topic1  topic2 topic3 topic4 topic5 topic6
## topic2 0.0042  -      -      -      -      -
## topic3 0.0259  0.7545 -      -      -      -
## topic4 0.0259  0.7817 0.9250 -      -      -
## topic5 8.2e-05 0.1491 0.0906 0.0906 -      -
## topic6 0.4568  0.0423 0.0906 0.0906 0.0012 -
## topic7 0.0906  0.4880 0.7054 0.7054 0.0522 0.3298
##
## P value adjustment method: BH
```

Old vs New Era: does sentiment differ?

```r
era_sentiment <- topic_source |>
  filter(!is.na(era))

cat("Sample sizes by era:\n")
```

```
## Sample sizes by era:
```

```
print(table(era_sentiment$era))
```

```
##
##       new_only references_old
##           1698             39
```

```
# Wilcoxon rank-sum (non-parametric, two groups)
if (length(unique(era_sentiment$era)) == 2 &&
    all(table(era_sentiment$era) > 5)) {
  era_test <- wilcox.test(sentiment ~ era, data = era_sentiment)
  print(era_test)
} else {
  cat("Not enough old-brand references for a reliable test.\n")
}
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  sentiment by era
## W = 29572, p-value = 0.2525
## alternative hypothesis: true location shift is not equal to 0
```

```
# Mean sentiment by era
era_sentiment |>
  group_by(era) |>
  summarise(mean_sentiment = mean(sentiment, na.rm = TRUE),
            n = n(), .groups = "drop") |>
  knitr::kable(digits = 3, caption = "Sentiment: old-brand references vs new-only")
```

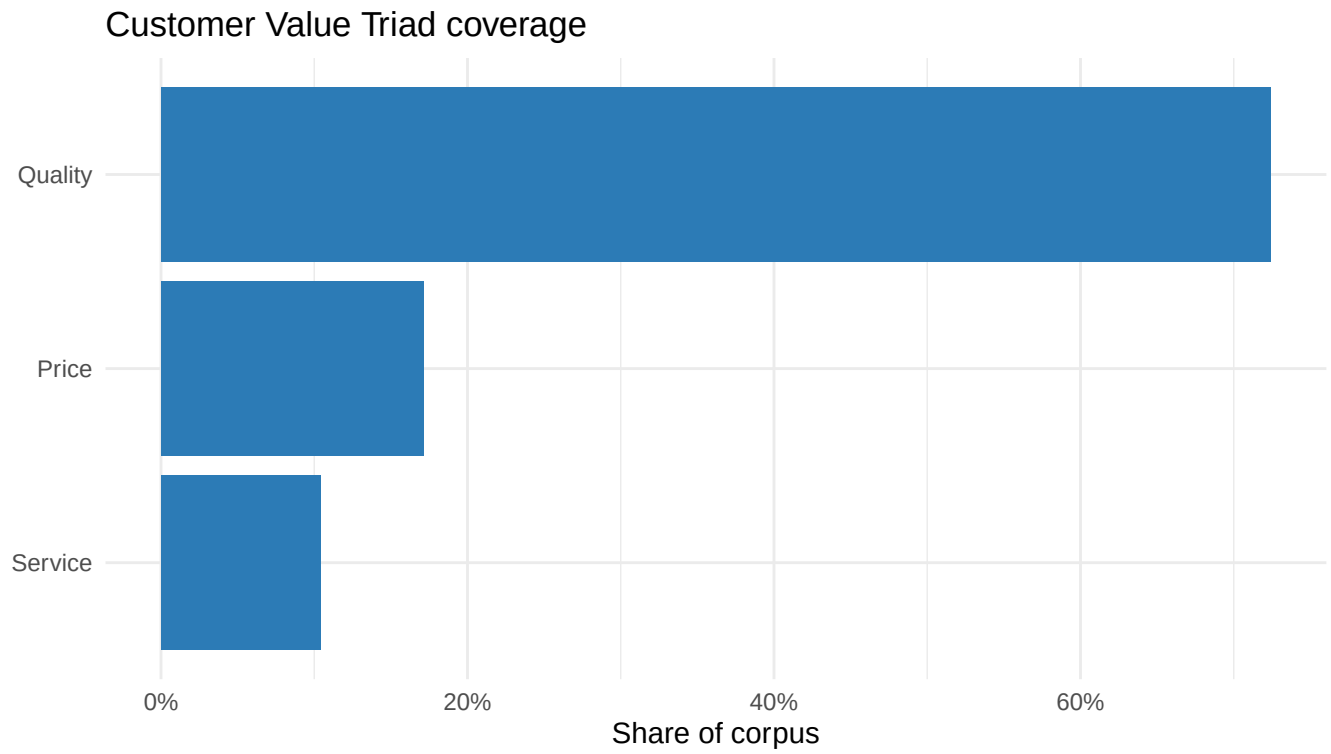
Table 7: Sentiment: old-brand references vs new-only

era	mean_sentiment	n
new_only	0.080	1698
references_old	0.107	39

Customer Value Triad

```
# REVIEW this mapping against your final topic labels.
triad_map <- data.frame(
  topic = paste0("topic", 1:7),
  dimension = c("Quality",   # Software & UI
                "Quality",   # Form Factor & Design
                "Quality",   # Competitor Comparison
                "Quality",   # Hardware & Specs
                "Price",     # Price & Purchase
                "Quality",   # Apps & Functionality
                "Service"))  # Ordering & Delivery
```

```r
triad_summary <- topic_counts |>
  mutate(topic = as.character(topic)) |>   # already "topic1", just coerce factor->char
  left_join(triad_map, by = "topic") |>
  group_by(dimension) |>
  summarise(total_prop = sum(prop), .groups = "drop")

ggplot(triad_summary, aes(x = reorder(dimension, total_prop), y = total_prop)) +
  geom_col(fill = "#2C7BB6") +
  coord_flip() +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Customer Value Triad coverage",
       x = NULL, y = "Share of corpus") +
  theme_minimal()
```




```r
knitr::kable(triad_summary, digits = 3,
             col.names = c("Value dimension", "Share of corpus"),
             caption = "Customer value triad reconstructed from topics")
```

Table 8: Customer value triad reconstructed from topics

Value dimension	Share of corpus
Price	0.172
Quality	0.724
Service	0.104

Reflection